TruSeq Index Primers

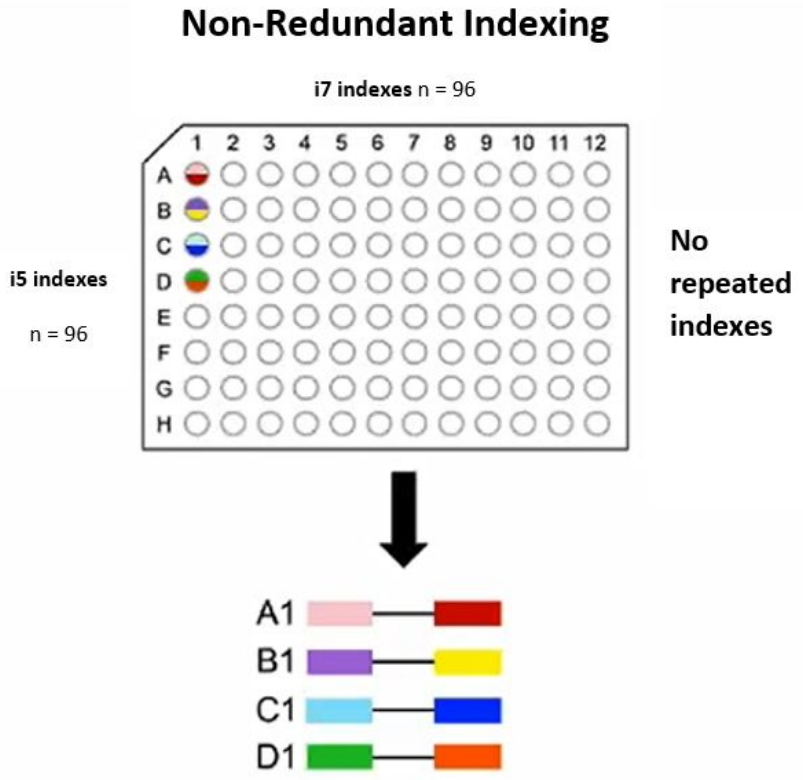
Biohub-designed TruSeq-compatible primers with 12bp indices for high index diversity

Background

- Motivation: With 4x96 of 8nt dual unique indexing primers, only 384 primer pairs are in use. This means that often the same indexes are used in back-to-back experiments, raising concern for same-index library contamination within the post-PCR lab or on the sequencer itself. This is, of course, a particular worry for pathogen detection.

- Solution: Purchase a set of 12x i7 and 12x i5 ninety-six-well plates of indexing primers to be used combinatorially for a total of 144 plates.

Order i7 and i5 plates to mix and make TruSeq, Dual-Indexed 12bp Barcode Plates



It is possible to make 144 combinations of Dual Unique Barcode Plates

- The full set of barcodes will include 144 plates with 96 wells each for a total of 13,824 barcodes
- All of the barcodes have a hamming distance of at least 3
- All i7 barcodes will be different from all i5 barcodes, ensuring that everything is NovaSeq compatible

12 dual unique i5 plates	x	12 dual unique i7 plates	=	144 dual unique i5/i7 mixed plates

12 x 12 = 144 dual barcode plates

Dual indexes are fully unique within each set of 12, but not between sets

Dual-001 = i7-01 + i5-01	Dual-025 = i7-01 + i5-03	Dual-049 = i7-01 + i5-05	Dual-073 = i7-01 + i5-07	Dual-097 = i7-01 + i5-09	Dual-121 = i7-01 + i5-11
Dual-002 = i7-02 + i5-02	Dual-026 = i7-02 + i5-04	Dual-050 = i7-02 + i5-06	Dual-074 = i7-02 + i5-08	Dual-098 = i7-02 + i5-10	Dual-122 = i7-02 + i5-12
Dual-003 = i7-03 + i5-03	Dual-027 = i7-03 + i5-05	Dual-051 = i7-03 + i5-07	Dual-075 = i7-03 + i5-09	Dual-099 = i7-03 + i5-11	Dual-123 = i7-03 + i5-01
Dual-004 = i7-04 + i5-04	Dual-028 = i7-04 + i5-06	Dual-052 = i7-04 + i5-08	Dual-076 = i7-04 + i5-10	Dual-100 = i7-04 + i5-12	Dual-124 = i7-04 + i5-02
Dual-005 = i7-05 + i5-05	Dual-029 = i7-05 + i5-07	Dual-053 = i7-05 + i5-09	Dual-077 = i7-05 + i5-11	Dual-101 = i7-05 + i5-01	Dual-125 = i7-05 + i5-03
Dual-006 = i7-06 + i5-06	Dual-030 = i7-06 + i5-08	Dual-054 = i7-06 + i5-10	Dual-078 = i7-06 + i5-12	Dual-102 = i7-06 + i5-02	Dual-126 = i7-06 + i5-04
Dual-007 = i7-07 + i5-07	Dual-031 = i7-07 + i5-09	Dual-055 = i7-07 + i5-11	Dual-079 = i7-07 + i5-01	Dual-103 = i7-07 + i5-03	Dual-127 = i7-07 + i5-05
Dual-008 = i7-08 + i5-08	Dual-032 = i7-08 + i5-10	Dual-056 = i7-08 + i5-12	Dual-080 = i7-08 + i5-02	Dual-104 = i7-08 + i5-04	Dual-128 = i7-08 + i5-06
Dual-009 = i7-09 + i5-09	Dual-033 = i7-09 + i5-11	Dual-057 = i7-09 + i5-01	Dual-081 = i7-09 + i5-03	Dual-105 = i7-09 + i5-05	Dual-129 = i7-09 + i5-07
Dual-010 = i7-10 + i5-10	Dual-034 = i7-10 + i5-12	Dual-058 = i7-10 + i5-02	Dual-082 = i7-10 + i5-04	Dual-106 = i7-10 + i5-06	Dual-130 = i7-10 + i5-08
Dual-011 = i7-11 + i5-11	Dual-035 = i7-11 + i5-01	Dual-059 = i7-11 + i5-03	Dual-083 = i7-11 + i5-05	Dual-107 = i7-11 + i5-07	Dual-131 = i7-11 + i5-09
Dual-012 = i7-12 + i5-12	Dual-036 = i7-12 + i5-02	Dual-060 = i7-12 + i5-04	Dual-084 = i7-12 + i5-06	Dual-108 = i7-12 + i5-08	Dual-132 = i7-12 + i5-10
Dual-013 = i7-01 + i5-02	Dual-037 = i7-01 + i5-04	Dual-061 = i7-01 + i5-06	Dual-085 = i7-01 + i5-08	Dual-109 = i7-01 + i5-10	Dual-133 = i7-01 + i5-12
Dual-014 = i7-02 + i5-03	Dual-038 = i7-02 + i5-05	Dual-062 = i7-02 + i5-07	Dual-086 = i7-02 + i5-09	Dual-110 = i7-02 + i5-11	Dual-134 = i7-02 + i5-01
Dual-015 = i7-03 + i5-04	Dual-039 = i7-03 + i5-06	Dual-063 = i7-03 + i5-08	Dual-087 = i7-03 + i5-10	Dual-111 = i7-03 + i5-12	Dual-135 = i7-03 + i5-02
Dual-016 = i7-04 + i5-05	Dual-040 = i7-04 + i5-07	Dual-064 = i7-04 + i5-09	Dual-088 = i7-04 + i5-11	Dual-112 = i7-04 + i5-01	Dual-136 = i7-04 + i5-03
Dual-017 = i7-05 + i5-06	Dual-041 = i7-05 + i5-08	Dual-065 = i7-05 + i5-10	Dual-089 = i7-05 + i5-12	Dual-113 = i7-05 + i5-02	Dual-137 = i7-05 + i5-04
Dual-018 = i7-06 + i5-07	Dual-042 = i7-06 + i5-09	Dual-066 = i7-06 + i5-11	Dual-090 = i7-06 + i5-01	Dual-114 = i7-06 + i5-03	Dual-138 = i7-06 + i5-05
Dual-019 = i7-07 + i5-08	Dual-043 = i7-07 + i5-10	Dual-067 = i7-07 + i5-12	Dual-091 = i7-07 + i5-02	Dual-115 = i7-07 + i5-04	Dual-139 = i7-07 + i5-06
Dual-020 = i7-08 + i5-09	Dual-044 = i7-08 + i5-11	Dual-068 = i7-08 + i5-01	Dual-092 = i7-08 + i5-03	Dual-116 = i7-08 + i5-05	Dual-140 = i7-08 + i5-07
Dual-021 = i7-09 + i5-10	Dual-045 = i7-09 + i5-12	Dual-069 = i7-09 + i5-02	Dual-093 = i7-09 + i5-04	Dual-117 = i7-09 + i5-06	Dual-141 = i7-09 + i5-08
Dual-022 = i7-10 + i5-11	Dual-046 = i7-10 + i5-01	Dual-070 = i7-10 + i5-03	Dual-094 = i7-10 + i5-05	Dual-118 = i7-10 + i5-07	Dual-142 = i7-10 + i5-09
Dual-023 = i7-11 + i5-12	Dual-047 = i7-11 + i5-02	Dual-071 = i7-11 + i5-04	Dual-095 = i7-11 + i5-06	Dual-119 = i7-11 + i5-08	Dual-143 = i7-11 + i5-10
Dual-024 = i7-12 + i5-01	Dual-048 = i7-12 + i5-03	Dual-072 = i7-12 + i5-05	Dual-096 = i7-12 + i5-07	Dual-120 = i7-12 + i5-09	Dual-144 = i7-12 + i5-11

Ordering

The sequences of the full primers to order can be found in:

Biohub_TruSeq_Index_Primer_FullSequence.csv

An index lookup template to assist in sample sheet preparation is also available:

Template_Biohub_TruSeq_Index_Lookup.xlsx